Tangents and normals

1 For each of the following curves and given point:

i Find the equation of the tangent. **ii** Find the equation of the normal.

a $f(x) = 2e^x$ at the point where $x = 0$.

b $f(x) = 5e^{4x}$ at the point $(0, 5)$.

c $f(x) = \dfrac{e^x}{6}$ at the point $\left(0, \dfrac{1}{6}\right)$.

d $f(x) = 4e^{3x} - 2x$ at the point where $x = 0$.

2 For each of the following curves, at the given point:

i Find the exact gradient of the normal. **ii** Find the equation of the normal.

a $y = \dfrac{e^x}{x}$ at the point where $x = -1$.

b $y = xe^x$ at the point where $x = 1$.

c $y = x - e^{-x}$ at the point where $x = 2$.

3 The tangent to the curve $f(x) = 2.5e^x$ at the point $x = 7.5$, is parallel to the tangent to the curve $g(x) = e^{2.5x}$ at the point where $x = k$.
Find the value of k.

4 Consider $y = xe^{-x}$:

a Find the gradient function.

b Find the gradient of the normal at the point $\left(3, 3e^{-3}\right)$.

c Hence, find the equation of the normal at the point $\left(3, 3e^{-3}\right)$.

5 The tangent to the curve $y = x^2e^x$ at $x = 1$, cuts the axes at points A and B .

a Find the exact gradient of the tangent at $x = 1$.

b Find the equation of the tangent at $x = 1$.

c Find the coordinates of point A, where the tangent cuts the x-axis

d Find the coordinates of point B, where the tangent cuts the y-axis

e Determine the exact area of triangle AOB where point O is the origin.

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6 Consider the functions $y = 3e^{-x}$ and $y = 2 + e^{-x}$:

- **a** Find the point of intersection of the functions.
- **b** Find the gradient of the tangent to $y = 3e^{-x}$ at the point of intersection.
- **c** Determine the acute angle this tangent makes with the x-axis, to the nearest degree.
- **d** Find the gradient of the tangent to $y = 2 + e^{-x}$ at the point of intersection.
- **e** Determine the acute angle this tangent makes with the x-axis, to the nearest degree.
- **f** Find the acute angle between the two tangents.

7 Consider the functions $g(x) = e^{7x}$ and $f(x) = e^{-7x}$.

- **a** Consider the tangent line to $g(x)$ at $x = 0$. Find the acute angle that the tangent line makes with the x-axis, correct to one decimal place.
- **b** Consider the tangent line to $f(x)$ at $x = 0$. Find the acute angle that the tangent line makes with the x-axis, correct to one decimal place.
- **c** Hence, determine the acute angle between the two tangents at $x = 0$.

8 Consider the curve $f(x) = e^x + ex$. Show that the tangent to the curve at the point $(1, 2e)$ passes through the origin.

Exponential graphs

9 Consider the function $f(x) = 3 - e^{-x}$.

- **a** Find $f(0)$.
- **b** Find $f'(0)$.
- **c** State whether $f(x)$ is an increasing or decreasing function. Explain your answer.
- **d** Find the limit of $f(x)$ as $x \to \infty$.

10 Consider the function $f(t) = \dfrac{4}{2 + 3e^{-t}}$.

- **a** Find $f(t)$ when $t = 0$.
- **b** Find $f'(t)$.
- **c** State whether $f(x)$ is an increasing or decreasing function. Explain your answer.
- **d** Hence, state how many stationary points the function has.
- **e** Find the limit of $f(t)$ as $t \to \infty$.

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11 Consider the function $y = e^x(x - 3)$.

a Find the coordinates of the turning point.

b State whether this is a minimum or maximum turning point.

12 Consider the function $f(x) = 4e^{-x^2}$.

a Find $f'(x)$.

b Find the values of x for which:

i $f'(x) = 0$ **ii** $f'(x) > 0$. **iii** $f'(x) < 0$

c Find the limit of $f(x)$ as $x \to \infty$.

d Find the limit of $f(x)$ as $x \to -\infty$.

e Sketch the graph of $f(x)$.

13 Consider the function $f(x) = e^x - 2e^{-x}$.

a Find the x-intercept, to two decimal places.

b Find the y-intercept.

c What value does y approach as $x \to \infty$.

d What value does y approach as $x \to -\infty$.

e Sketch the graph of the function.

14 For each of the following functions:

i Find the x-intercept.

ii Find the y-intercept.

iii Find the coordinates of any stationary points.

iv State the nature of the stationary points.

v Find any points of inflection.

vi Describe what happens to the function as $x \to \infty$.

vii Sketch the graph of the function.

a $y = xe^{-x}$ **b** $y = e^{2x} - 3e^x$

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15 Consider the function $y = x^3 e^{-x}$:

- **a** Find the x-intercept.
- **b** Find the y-intercept.
- **c** Find y'.
- **d** Find the coordinates of any stationary points, correct to two decimal places if necessary.
- **e** Describe the nature of the stationary points.
- **f** Sketch the graph of the function.

16 For the function $y = e^{-x}\sqrt{x}$:

- **a** Find the x-intercept.
- **b** Find the y-intercept.
- **c** Find y'.
- **d** Find the coordinates of any stationary points, correct to two decimal places.
- **e** State the nature of the stationary points.
- **f** Sketch the graph of the function.

17 Consider the function $y = e^{2x} - 3e^{x}$.

- **a** Find the x-intercept.
- **b** Find the y-intercept.
- **c** Find the coordinates of any stationary points, correct to two decimal places.
- **d** State the nature of the stationary point.
- **e** Describe the behaviour of this function as $x \to \infty$ and $x \to -\infty$.
- **f** Sketch the graph of the function.

Exponential growth and decay

18 The local rodent population, numbering approximately 840, is decreasing at a rate of 4% per year.

- **a** Write a function, y, to represent the population after m years.
- **b** Find the size of the population after 8 years, giving your answer to the nearest whole number.

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19 Switzerland's population in the next 10 years is expected to grow approximately according to the model $P = 8\,(1 + r)^t$, where P represents the population (in millions), t years from now.
The world population in the next 10 years is expected to grow approximately according to the model $Q = 7130\,(1.0133)^t$, where Q represents the world population (in millions), t years from now.

a Use the model for P to find the current population in Switzerland.

b Use the model for Q to find:

i The current world population

ii The population of the world in twenty years time, to the nearest million.

20 The population of China (in millions), t years after 1980, is modelled by $C = 991e^{0.011t}$.
The population of India (in millions), t years after 1980 is modelled by $I = 680e^{0.024t}$.

a After how many years will India's population first exceed China's population. Round your answer to the nearest whole number.

b Find the time, t, at which the instantaneous growth rate of India's population is double that of China. Round your answer to one decimal place.
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21 The area covered by an ice shelf was measured over some warmer months. At the beginning of the first month the ice shelf was spread over an area of $1437\ \text{km}^2$, and it was found that this area decreased by 2% each month over the recording period.

a Find the area that was covered by the ice shelf after 13 months of recording. Round your answer to the nearest square kilometre.

b Hence, find the loss in the ice shelf's area over 13 months.

22 A property has an original value of P and doubles in value every 14 years. Find the annual rate of growth, as a percentage to two decimal places.

23 The concentration, x, of a certain medication in the bloodstream of a patient, t hours after taking a dose, is given by $x = 8te^{-4t}$.

a State the concentration in the bloodstream when the patient initially takes the dose.

b Find the time, t, at which the patient has the greatest concentration of medication in his bloodstream.

c Find the maximum concentration of the medication. Round your answer to three decimal places.

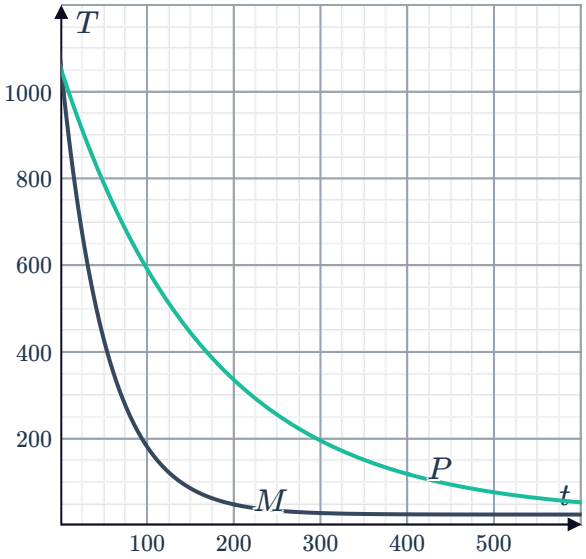
d According to the model, does the concentration ever reach 0 again?

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24 Consider two metals M and P that are heated to a temperature of 1050° C. They are both placed in the same room where the temperature is a constant 25° C, so that the rate of cooling for each metal can be given by

$$\frac{dT}{dt} = -k\,(T - 25)$$

and the model for temperature T at time t can be given by $T = 25 + 1025e^{-kt}$.
The temperature of each metal over time is shown in the graph:



a State which metal cools more quickly.

b State which parameter, k or T, accounts for the difference in the rate of cooling of the two metals.

25 When an object is heated to a certain temperature and then placed in an environment of constant 25° C temperature, it starts to cool. The temperature of the object T after t minutes of cooling is given by:

$$T = 25 + 1010e^{-0.002t}$$

The rate at which it cools is given by:

$$\frac{dT}{dt} = -0.002\,(T - 25)$$

a State the initial temperature of the object when it is placed in the environment of constant temperature.

b If the object is never removed from that environment, state what temperature will it eventually approach.

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26 A pastry chef has perfected a cake recipe where the cake bakes to a temperature of $165^\circ\ \mathrm{C}$ in the oven, and is then immediately removed and placed in a freezer of constant temperature $-5^\circ\ \mathrm{C}$. After t minutes in the freezer, the temperature $T^\circ\ \mathrm{C}$ at the centre of the cake is given by:

$$T = A - Be^{-0.03t}$$

a Find the values of A and B.

b When the temperature of the centre of the cake drops to $14^\circ\ \mathrm{C}$, the chef can remove the cake from its tin without it breaking apart. Find the number of minutes, t, the cake needs to remain in the freezer before the chef can remove it from its tin. Round your answer to the nearest minute.

c Find the rate at which the cake cools when it is removed from the fridge, correct to one decimal place.

27 An economist expects that over the next 12 years, the value of the US dollar (relative to the Australian dollar) will change according to the formula:

$$f\left(t\right) = \frac{127\left(92^t\right)}{10^{2(t+1)}}$$

where t represents the number of years from now.

a Express the formula in the form $f\left(t\right) = A(B)^t$, where A and B are decimals.

b By what percentage is the value of the US dollar expected to decrease each year?

c Find the value of the US dollar, relative to the Australian dollar in 12 years time.

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